Rohan Hitchcock

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Experience

Research Scientist Jul. 2026 – Present
Resolution

Research Scientist Mar. 2026 – Jul. 2026
Timaeus

Research Affiliate (while completing PhD) Oct. 2025 – Feb. 2026
Timaeus

Postgraduate Student Jan. 2023 – Feb. 2026
The University of Melbourne & CSIRO

Tutor (Teaching Assistant) Jul. 2020 – Jun. 2025
The University of Melbourne

Research Assistant Jun. 2019 – Aug. 2020
The Peter Doherty Institute for Infection and Immunity

Research Intern Jun. 2018 – Jun. 2019
The Peter Doherty Institute for Infection and Immunity

Education

Doctor of Philosophy Jan. 2023 – Feb. 2026
The University of Melbourne & CSIRO

Master of Science (with Distinction), Mathematics and Statistics Dec. 2022
The University of Melbourne

Bachelor of Science, Mathematics and Statistics Jul. 2020
The University of Melbourne

Diploma in Computing Jul. 2020
The University of Melbourne

Publications

- **R. Hitchcock** (2026) "Singular Learning Theory for Deep Learning Interpretability: Stochastic Sampling Methods and Applications to Learning from Physical Systems," *PhD Thesis*, The University of Melbourne.
- **R. Hitchcock**, G. W. Delaney, J. H. Manton, R. Scalzo and J. Zhu (2026) "Susceptibilities for Neural Networks Learning from Physical Data," *Advances in Neural Information Processing Systems (NeurIPS 2026)*.
- **R. Hitchcock** and J. Hoogland (2025) "From Global to Local: A Scalable Benchmark for Local Posterior Sampling," arXiv:2507.21449.
- **R. Hitchcock**, G. W. Delaney, J. H. Manton, R. Scalzo and J. Zhu (2025) "Emergence of Computational Structure in a Neural Network Physics Simulator," *XAI4Science Workshop at the International Conference on Learning Representations (ICLR 2025)*.
- **R. Hitchcock** (2022) "Differentiation, Division and the Bicategory of Landau–Ginzburg Models," *MSc Thesis*, The University of Melbourne.
- Y. Kato, T. M. Steiner, H.-Y. Park, **R. Hitchcock** et al. (2020) "Display of Native Antigen on cDC1 That Have Spatial Access to Both T and B Cells Underlies Efficient Humoral Vaccination," *Journal of Immunology* 205(7):1842–1856.

Research Notes and Talks

- A. Gordon, **R. Hitchcock**, I. Mouratidis, J. Wind and D. Murfet (2026) "Interpretability as a Benchmark for Sampling," *Resolution*.
- **R. Hitchcock**, J. Hoogland, G. Wang and A. Gordon (2026) "Guide for Sampling Hyperparameter Selection," *Timaeus*.
- A. Gordon, **R. Hitchcock** and D. Murfet (2026) "Interpreting the Ising Model," *Timaeus*.
- D. Murfet, A. Gordon, M. Adam, G. Wang, J. Hoogland, G. Baker, W. Snell, S. van Wingerden, A. Newgas, B. Snikkers and **R. Hitchcock** (2026) "Spectroscopy at Scale: Finding Interpretable Structure in Pythia-1.4B," *Timaeus*.
- "Sampling Hyperparameter Selection in SLT," talk, ILIAD 2026, Berkeley.
- "Towards Higher Confidence in Alignment," talk, Australian AI Safety Forum 2026, Sydney.
- "The Empirical Process in Singular Learning Theory," talk, SLT Seminar 2025, online.
- "Local Learning Coefficient Estimation: The Story So Far and Open Problems," talk, ILIAD 2025, Berkeley.
- "Induction Heads and Phase Transitions," talk, SLT & Alignment Summit 2023, Berkeley.